

Robayed Ashraf

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SUMMARY

AI/ML engineer who built DocuMind, a multi-agent RAG system with its own MCP server, from the ground up. A LangGraph supervisor of Researcher, Synthesizer, and Critic agents catches hallucinations before a user ever sees the answer, scoring 0.985 faithfulness on RAGAS. In addition, built SignSync, a real-time sign language translator for video conferencing running at 99.79% accuracy in live video. Owns the full stack: architecture, retrieval, deployment, and CI/CD.

TECHNICAL SKILLS

AI & Machine Learning: Agentic AI, Multi-Agent Systems, LLM Applications, RAG, MCP (Model Context Protocol), Tool Calling, Generative AI, Deep Learning, Computer Vision, NLP, Prompt Engineering, Hybrid Search, Vector Embeddings, Contextual Retrieval, LLM Evaluation (RAGAS)

Frameworks: Python, PyTorch, TensorFlow, Keras, LangChain, LangGraph, FastAPI, OpenCV, Scikit-Learn, HuggingFace, Redis, Celery

MLOps & DevOps: Docker, Docker Compose, GitHub Actions (CI/CD), ClearML, Pytest, REST API Development

Tools & Practices: Git, SQL, PostgreSQL, Agile/Scrum, Jira, Confluence

PROJECTS

DocuMind: Multi-Agentic Document Intelligence

Feb 26 – Present

Tech: LangGraph, LangChain, Gemini 2.5 Flash, MCP, FastAPI, Next.js 16, Postgres, pgvector, Redis, Celery, Tavily, HuggingFace, Docker, GitHub Actions, RAGAS, Pytest

- Multi-agent RAG system for chatting with PDFs and DOCX: a LangGraph supervisor runs Researcher, Synthesizer, and Critic agents with self-correcting revision loops.
- Agentic tool use via Gemini function calling, with the Researcher autonomously invoking web search and a sandboxed calculator.
- Hybrid retrieval with pgvector HNSW (dense) and ts_rank (sparse) search, fused via RRF with cross-encoder reranking, Contextual Retrieval, and HyDE fallback.
- Scores: 0.985 faithfulness and 0.917 context recall on a 30-question RAGAS golden set, deployed via Docker Compose with GitHub Actions CI/CD.

SignSync: Sign Language Translation for Video Conferencing

Feb 25 – Jun 25

Tech: Python, ANN, OpenCV, Gemini API (LLM integration), Computer Vision

- Trained an ANN on 3,200 images across 42 classes for real-time AUSLAN finger spelling classification, achieving 99.79% validation accuracy.
- Integrated Gemini API for LLM-based post-processing, correcting recognition errors, and improving caption readability in real time.
- Designed a 30-frame confidence-based smoothing mechanism to stabilize live predictions in continuous video streams.
- Deployed the full pipeline into a live Jitsi environment, delivering real-time sign-to-text captions as a production-viable accessibility feature.

ThirdAxis: Smart Plant Health Monitor

Feb 25 – Jun 25

Tech: Python, CNN, ClearML, Streamlit, GitHub Actions, Jira, Confluence

- Managed end-to-end ML project lifecycle using Agile practices with Jira and Confluence, covering data ingestion, preprocessing, training, evaluation, and deployment.
- Designed a modular ClearML pipeline with separate scripts for each stage, enabling reproducible and traceable experimentation.
- Implemented ClearML artifact management for dataset and model versioning, supporting experiment comparison and rollback across training iterations.
- Built a Streamlit inference interface for real-time plant disease classification from leaf images, addressing early-detection needs in precision agriculture.

Multi-Object Tracking System for Real-Time Video Inference

Jan 25 – Jun 25

Tech: PyTorch, OpenCV, YOLOX, MOTR, TrackEval, CUDA, Streamlit

- Compared MOTR (Transformer-based) and BoostTrack++ (tracking-by-detection with YOLOX) on MOT17, with MOTR achieving 98.518% IDF1 and 92.57% HOTA versus 96.104% and 87.093% for BoostTrack++.
- Profiled real-time inference on NVIDIA RTX 4060, recording 1.54 to 3.60 FPS and 255 to 409ms latency to identify deployment trade-offs for live video scenarios.
- Built a modular Streamlit prototype supporting live webcam and uploaded video inference with real-time bounding box overlays, FPS display, and latency tracking.
- Evaluated tracking quality using TrackEval metrics, including MOTA, IDF1, and HOTA, against MOT17 ground truth annotations.

WORK EXPERIENCE

Self-Employed, AI/ML Engineer

Feb 26 – Present

- Sole engineer across three major releases, owning all decisions from retrieval design and agent orchestration to database schema and deployment.
- Re-architected the linear RAG pipeline into a multi-agent LangGraph supervisor, gating the redesign behind RAGAS regression checks before shipping.
- Built the full backend: FastAPI REST API, Celery worker queue, Postgres with pgvector, Redis caching and rate limiting, Clerk JWT auth, and an MCP server for Claude Desktop and Cursor.
- Shipped a production Next.js 16 frontend with SSE streaming chat, inline PDF citations, and per-user document isolation.

East West University Computer Programming Club, Programming Trainer (Volunteering)

Jan 19 – Oct 21

- Mentored 50+ students in algorithms, data structures, and competitive programming.
- Designed and conducted ICPC-style mock contests to build structured problem-solving skills.

EDUCATION

University of Technology Sydney, Master of Artificial Intelligence (Major: Computer Vision)

Aug 23 – Jun 25

WAM: 80.94 / 100 (Grade: 6.13 / 7.00)

- Selected and presented the SignSync Project at UTS Tech Fest AI Showcase 2025, recognized for practical impact among postgraduate AI projects.

East West University, Bachelor of Science in Computer Science and Engineering

Jan 18 – Oct 21

CGPA: 3.64 / 4.00

- Dean's List recipient and merit-based scholarship awardee for academic excellence.